Conceptual Framework & Multi-Domain Applications Guide

Proposed Cognitive Graph Digital Twin (GDT) Adaptive Cyber-Security Architecture

Anshika Keshari[*1], **Mallikharjuna Rao K**[2], **Sabarathinam Chockalingam**[3], **Sandeep Pirbh**

[1]KIIT Bhubaneswar & IIIT Naya Raipur, [2]IIIT Naya Raipur, [3]IFE Norway, [4]NR Norway

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Overview of the Proposed Conceptual Architecture

The proposed **Cognitive Graph Digital Twin (GDT) Adaptive Cyber-Security Framework** operates across three interconnected pillars:

1. **Physical Cyber-Physical Twin (IT-OT Domain)**: Comprises SCADA control centers, industrial 5G networks, and field Operational Technology (OT) assets.
2. **Virtual Graph Digital Twin Layer**: Real-time spatial graph mirror $G(V, E)$ tracking nodal features (V, θ, P, Q) and admittance attributes (Y_{bus}).
3. **Cognitive Decision-Making & Healing Cycle**: Four-stage Observe-Learn-Plan-Act (O-L-P-A) loop executing Multi-Task GraphSAGE detection, Integrated Gradients explainability, and closed-loop self-healing.

Detailed Breakdown of the Three Core Pillars

Physical Cyber-Physical Twin (IT-OT Domain)

- **IT Layer**: SCADA master stations, cloud management servers, and energy management systems (EMS).
- **Industrial Network Layer**: High-speed communication protocols (5G slices, IEC 61850) streaming sensor telemetry.
- **OT Field Layer**: Physical hardware assets including Remote Terminal Units (RTUs), Phasor Measurement Units (PMUs), busbar substations, and circuit breakers.

Virtual Graph Digital Twin Layer

Spatial admittance graph representation $G = (V, E, X, E_{\text{attr}})$ where buses are nodes, transmission lines are edges, and nodal admittance $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}}$ forms the graph topology. Real-time synchronization allows the virtual graph to dynamically update its edge connections when lines trip ($N-1$ outages).

Cognitive Decision-Making & Adaptive Security Cycle (O-L-P-A)

1. **Observe & Orient**: Normalizes raw SCADA sensor streams (V, θ, P, Q) and constructs spatial PyTorch Geometric graph objects.
2. **Learn**: Multi-task GraphSAGE GNN backbone executes concatenation-based message passing:
$$\mathbf{h}_v^{(k)} = \text{ReLU}\left(\mathbf{W}_k \cdot \left[\mathbf{h}_v^{(k-1)} \,\|\, \mathbf{h}_{\mathcal{N}(v)}^{(k)}\right]\right) \tag{1}$$
Predicts graph-level attack presence (98.25% F1) and node anomalies (97.62% F1) in 0.908 ms.
3. **Plan & Explain**: Integrated Gradients (IG) computes exact per-sensor attributions $(\sum \text{Attr}_i = \Delta F)$.
4. **Decide & Act**: Quarantines corrupted RTU channels and substitutes load baseline forecasts ($\pm 2\%$ noise), reducing state estimation MSE from 0.007228 p.u.^{2} down to 0.000002 p.u.^{2} (**99.97% error recovery**).

Multi-Domain CPS Applications

While our paper focuses on **Smart Power Grids (IEEE 14-bus / 30-bus)**, the Graph Digital Twin framework is domain-agnostic:

CPS Application	**Graph Nodes (V)**	**Graph Edges (E)**	**Telemetry Signals (X)**
Smart Power Grids	Transmission Substations	Power Lines	Voltage (V), Angle (θ), Power (P, Q)
Water Distribution Networks	Pipe Junctions & Pumps	Water Main Pipes	Water Flow Rate, Hydraulic Pressure
Industrial Microgrids	Battery / Solar Inverters	Microgrid Busbars	State-of-Charge (SoC), Frequency, Power
Smart EV Charging Grids	Fast Charging Depots	Substation Feeders	Charging Demand, Transformer Voltage